De La Salle Holy Cross College

NOVEMBER EXAMINATION

2023

Grade 10

LIFE SCIENCES

Practical Examination

Examiner: Mrs G. Tolley **Time:** 90 minutes

Moderators: Mrs D. Webb, Ms B. Hannweg **Marks:** 50

Name: _____ **Class**_____

PLEASE READ THROUGH THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of 8 pages and an information sheet of one page. Please check that your question paper is complete.
2. You have 10 minutes to read through the paper before you begin. You are advised to read carefully and spend time planning your work.
3. Perform the tasks with care. You will be assessed on your ability to work neatly and independently.
4. Answer ALL the questions on this question paper in the space provided.
5. The information sheet is printed on coloured paper. Please read it carefully before you begin and refer to it during the course of the examination.

Part One – Conducting the task	Possible mark	Actual mark
Manipulation	5	
Questions 10 to 13	22	
Part Two – New method	23	
Total	**50**	

Remember that God is with you and will help and inspire you in all that you do

CRITERIA		
Sufficient care taken to prevent cross-contamination of powders on tile/petri dish	1	0
Iodine solution contained in each sample and not running between samples	1	0
Same amount of solution in all three test tubes	1	0
Test tubes correctly labelled	1	0
Working independently	1	0
TOTAL **MAX 5**		

PART ONE

The South African Police discovered a small amount of white powder scattered on the ground besides a victim they believe may have been murdered. The victim was found lying on the floor in his bedroom with a gag in his mouth and his hands and feet were bound so that he could not move. They have determined that the victim suffered from Type 1 diabetes and had to inject himself regularly with insulin in order to control his blood sugar level. In the kitchen of his home they found a new supply of insulin as well as several syringes and needles.

Is this a case of foul play?

During this examination you will be assuming the role of the forensic scientist to determine whether the unknown white powder found next to the dead victim had anything to do with his death.

You will carry out a short investigation to find out how certain white powders respond when treated with certain chemicals.

a. Before you begin with your investigation, make sure you have the following equipment at your workstation:
    - 3 test tubes in a test tube rack
    - Beaker
    - Table salt (A)
    - Corn starch (B)
    - Unknown white powder from the crime scene (C)
    - Ice cream stick
    - White tile
    - Piece of paper towel
    - 5ml syringe
    - Benedict's solution (at a central station)
    - Iodine solution (at a central station)
    - Dropper/pipette (at a central station)
    - 100ml cold tap water (H_2O)
    - Test tube holder
    - Permanent marker

b. You have access to boiling water,(water bath),at the central station

c. **NB. DONOT ATTEMPT TO TASTE ANY OF THE WHITE POWDERS**

TASK ONE

1. Place a small amount (ice cream stick tip) of about 1cm^3 of each white powder into a clean test tube and label the test tubes with a permanent marker as follows:
    - A (table salt)
    - B (corn starch)
    - C (unknown white powder from the crime scene)

2. Using the syringe, measure 2ml of cold water into each of the three test tubes.

3. Shake gently to dissolve the powder if possible.

4. Now, using the syringe, measure 2ml of Benedict's solution into each test tube and shake gently to mix thoroughly.

5. Place your test tubes in the beaker of hot water you have collected from the central station hot water bath for at least 5 minutes.

6. While you are waiting for the test tubes, place a small amount (ice cream stick tip) of about 1cm^3 of each of the white powders onto the tile. Make sure the powder samples remain separated from each other. Use the marking pen to provide labels A, B and C next to each sample.

7. Using the dropper/pipette, place 2 – 3 drops of iodine solution directly onto the white powder. Make sure the iodine does not run between the samples. **Observe any change in colour** of the iodine solution and make note of any colour changes that have occurred.

8. Using the test tube holder, carefully remove your test tubes from the hot water bath, place them in the test tube rack and return to your workstation.

9. **Observe** any **colour change** in the Benedict's solution or any precipitate found at the bottom of the test tubes.

NOW CALL THE TEACHER BEFORE YOU PROCEED ANY FURTHER.

10. Once your teacher has checked your work, draw a table in the space below in which you can record your results for each of the tests you have performed on the three samples of white powders. (6)

    The tests were:

    Benedict's test for a reducing sugar.

    Iodine test for starch.

    Record any **colour changes** you observed. (6)

    Give the table a heading.

Answer the following questions.

11. What conclusions can you draw about the unknown white powder found at the crime scene? (2)

12. When carrying out the investigation, describe how you worked carefully with the dropper/pipette and syringe in order to get results that are as accurate as possible. (4)

13. How could the design of the investigation done be improved?
Explain TWO improvements to the design. (4)

PART TWO

TASK TWO

You are now going to design a completely new experiment.

Design a **NEW EXPRIMENT** in which you determine **how effective the Benedict's solution is at detecting the presence of glucose in a solution at different temperatures.**

You may use any equipment found in your school laboratory.

MAKE SURE YOU HAVE READ THE INFORMATION SHEET CAREFULLY BEFORE YOU CARRY ON.

14.1. State your hypothesis. (3)

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14.2. State your aim. (2)

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14.3. Identify the independent variable. (2)

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14.4. Identify the dependent variable. (2)

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14.5. List any THREE controlled/fixed variables and state precisely how they will be controlled in your experiment to ensure fair testing. (6)

14.6. Outline your own NEW method. (8)

Modified from the NATIONAL SENIOR CERTIFICATE PAPER III 2011

TOTAL MARKS 50

INFORMATION SHEET
CRIME SCENE INVESTIGATION

A forensic scientist is a scientist who analyses material (biological, chemical or physical samples) found at a crime scene which can be used as evidence in a court of law. This work will provide the proof that the police need to bring a case to court. The three main branches of forensic science can be classified as biology, chemistry and drugs/toxicology.

Type I Diabetes is a disease which is characterised by the impaired (poor) production of insulin in the body. This results in high concentrations of glucose in the blood. The patient will suffer from dehydration because more water is found in the urine. The treatment of diabetes is very effective and requires the patient to inject themselves regularly (at least daily) with the hormone insulin. These insulin injections result in a lowering of blood sugar. Complex carbohydrates, e.g. starch have a lower glycaemic index (GI) and therefore do not cause problems for diabetics. Left untreated, the diabetic sufferer will ultimately fall into a coma which terminates in death.

The Benedict's test for glucose (reducing sugars) is a **quantitative test** used to determine the presence of glucose in solution.

- Place 2 ml of Benedict's solution into a test tube containing your sample.
- The Benedict's solution should remain a clear blue colour.
- Place the test tube in a boiling water waterbath for at least 5 minutes.
- Examine the sample after 5 minutes.
- An unchanged clear blue colour indicates no glucose present. A green precipitate* indicates a very small amount of glucose present. A yellow precipitate indicates a limited amount of glucose present. An orange-red precipitate indicates large amounts of glucose present.

**precipitate = an insoluble solid that emerges from a liquid solution and collects on the bottom of a test tube.*

The Iodine test for starch is a **qualitative test** which only indicates the presence or absence of starch in a sample. This simple test uses a prepared solution of iodine which is orange/brown in colour. The iodine solution turns blue/black in colour when it comes in contact with a sample containing starch.

A qualitative test is one which indicates the presence or absence of a particular substance in a sample.
A quantitative test is one which indicates the actual amount of a particular substance which is present in a sample. Quantitative tests range from extremely accurate and sensitive to very crude, (not sensitive).